ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Brief Introduction to Quantum Mechanics II

I. What is Hilbert Space?

II. What is the Hamiltonian of the system?

III. What is the eigenstate and eigenvalue?

IV. What basis is used for the wavefunction expansion?

I. What is Hilbert Space?

Hilbert space — a complete linear vector space over the complex field, equipped with an inner product

- **A complete and orthogonal eigen state composes a Hilbert space**
- **All wave functions are spanned within the Hilbert space**
- **Hilbert space consists of a finite or countably infinite dimension**

Eigenstate is discrete; rather than continuous

Eigenstate itself is infinite

Notion of 'Quantum' in quantum mechanics

$$\Psi = \begin{pmatrix} c_1|b_{11}\rangle & \cdots & c_n|b_{1n}\rangle \\ \vdots & \cdots & \vdots \\ c_1|b_{n1}\rangle & \cdots & c_n|b_{nn}\rangle \end{pmatrix}$$

***n* is infinite**

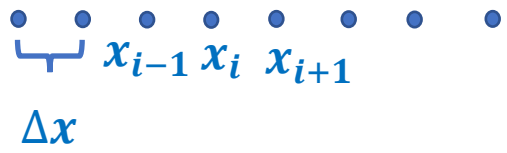
II. What is the Hamiltonian of the system?

Hamiltonian ($\hat{H}$) describes the total energy of the system

Kinetic Energy + Potential Energy

Single particle $\hat{H} = \frac{\hat{p}^2}{2m} + V(r) = \frac{(-i\hbar\nabla)^2}{2m} + V(r) = -\frac{\hbar^2\nabla^2}{2m} + V(r)$ **Schrödinger Equation**

Suppose space is discretized as a set of points in 1 dimension case

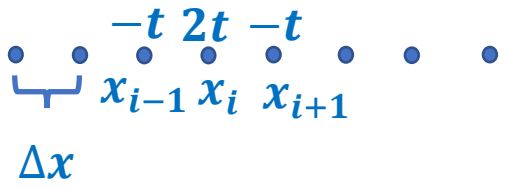


$$\nabla^2 = \frac{\partial^2}{\partial^2 x} \approx \left(\frac{x_{i+1} - x_i}{\Delta x} - \frac{x_i - x_{i-1}}{\Delta x}\right) \Big/ \Delta x = \frac{x_{i+1} - 2x_i + x_{i-1}}{(\Delta x)^2} = -\frac{-x_{i+1} + 2x_i - x_{i-1}}{(\Delta x)^2}$$

II. What is the Hamiltonian of the system?

Kinetic Energy $\hat{T} = -\frac{\hbar^2}{2m} \times -\frac{-x_{i+1} + 2x_i - x_{i-1}}{(\Delta x)^2} = \frac{\hbar^2}{2m(\Delta x)^2} \times -x_{i+1} + \frac{\hbar^2}{2m(\Delta x)^2} \times 2x_i + \frac{\hbar^2}{2m(\Delta x)^2} \times -x_{i-1}$

$$t = \frac{\hbar^2}{2m(\Delta x)^2}$$

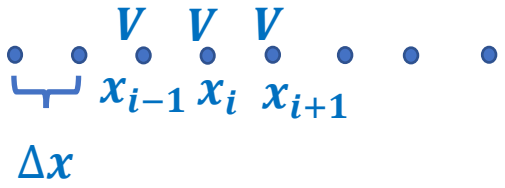


Matrix Form

$$\hat{T} = \begin{vmatrix} 2t & -t & 0 & 0 & \cdots & 0 \\ -t & 2t & -t & 0 & \cdots & 0 \\ 0 & -t & 2t & -t & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & 0 & -t & 2t & -t \end{vmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ \\ \vdots \\ x_n \end{matrix}$$

II. What is the Hamiltonian of the system?

Potential Energy $\hat{V}$



Matrix Form $\hat{V} =$

$$\begin{vmatrix} V(x_1) & 0 & 0 & 0 & \cdots & 0 \\ 0 & V(x_2) & 0 & 0 & \cdots & 0 \\ 0 & 0 & V(x_3) & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & 0 & 0 & 0 & V(x_3) \end{vmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ \\ x_n \end{matrix}$$

Matrix Form $\hat{H} = \hat{T} + \hat{V}$ $\hat{H} =$

$$\begin{vmatrix} 2t+V & -t & 0 & 0 & \cdots & 0 \\ -t & 2t+V & -t & 0 & \cdots & 0 \\ 0 & -t & 2t+V & -t & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & 0 & -t & 2t & -t+V \end{vmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ \\ x_n \end{matrix}$$

III. What is the eigenstate and eigenvalue?

Usually, the wavefunction, with an operator applied onto, turns into another different wavefunction

$$\hat{H}\Psi = \varphi$$

different from each other

Geometry view: **Operator, $\hat{A}$, changes both length and direction of wavefunction within the Hilbert space**

Physics view: **Wavefunction is the superposition of other state and collapses when a operator applied onto it**

Measurement of the observable under such state could give different values based on different probability

III. What is the eigenstate and eigenvalue?

Eigenstate — Some special wavefunctions, with an operator applied onto, maintain itself

$$\hat{H}\Psi = a\Psi$$

same from each other

Geometry view: **Operator, $\hat{A}$, only changes the length of such wavefunction; not the direction, within the Hilbert space**

Physics view: **Such wavefunction does not collapse when a operator applied onto it**

Measurement of the observable under such state could give fixed value with 100% probability

III. What is the eigenstate and eigenvalue?

How to compute the eigenstate of the system

Taking the Hamiltonian operator as an example $\hat{H}|\boldsymbol{\phi}_i\rangle = \begin{pmatrix} H_{11}\cdots H_{1n} \\ \vdots \quad \cdots \quad \vdots \\ H_{n1}\cdots H_{nn} \end{pmatrix} \begin{pmatrix} \boldsymbol{\phi}_1 \\ \vdots \\ \boldsymbol{\phi}_n \end{pmatrix} = a|\boldsymbol{\phi}_i\rangle = a \begin{pmatrix} \boldsymbol{\phi}_1 \\ \vdots \\ \boldsymbol{\phi}_n \end{pmatrix}$

$$\hat{H}|\boldsymbol{\phi}_i\rangle - a|\boldsymbol{\phi}_i\rangle = \begin{pmatrix} H_{11}\cdots H_{1n} \\ \vdots \quad \cdots \quad \vdots \\ H_{n1}\cdots H_{nn} \end{pmatrix} \begin{pmatrix} \boldsymbol{\phi}_1 \\ \vdots \\ \boldsymbol{\phi}_n \end{pmatrix} - a \begin{pmatrix} \boldsymbol{\phi}_1 \\ \vdots \\ \boldsymbol{\phi}_n \end{pmatrix} = \begin{pmatrix} H_{11}-a\cdots \quad H_{1n} \\ \vdots \quad \cdots \quad \vdots \\ H_{n1} \quad \cdots H_{nn}-a \end{pmatrix} \begin{pmatrix} \boldsymbol{\phi}_1 \\ \vdots \\ \boldsymbol{\phi}_n \end{pmatrix} = 0$$

The eigenstate $\begin{pmatrix} \boldsymbol{\phi}_1 \\ \vdots \\ \boldsymbol{\phi}_n \end{pmatrix}$ **cannot be zero so the determinant of** $\begin{pmatrix} H_{11}-a\cdots \quad H_{1n} \\ \vdots \quad \cdots \quad \vdots \\ H_{n1} \quad \cdots H_{nn}-a \end{pmatrix}$ **must be zero**

III. What is the eigenstate and eigenvalue?

According to linear algebra, $det\begin{pmatrix} H_{11}-a\cdots & H_{1n} \\ \vdots & \cdots \\ H_{n1} & \cdots H_{nn}-a \end{pmatrix}=0$**, means diagonalising the matrix**

$\hat{\boldsymbol{O}}$

Cramer's rule tells $det\hat{\boldsymbol{O}}=\hat{\boldsymbol{O}}\times Adj(\hat{\boldsymbol{O}})=0$

Do the calculation manually **or** **Call diagonalization function in LAPACK**

both a **and** $\boldsymbol{\phi}_i$ **can be obtained**

III. What is the eigenstate and eigenvalue?

Simple example: 2 energy level system $\hat{H} = \begin{pmatrix} \epsilon & \Delta \\ \Delta & \epsilon \end{pmatrix}$

Determinant: $det\begin{pmatrix} \epsilon - E & \Delta \\ \Delta & \epsilon - E \end{pmatrix} = 0$

Do the calculation manually

$$(\epsilon - E)^2 - \Delta^2 = 0 \begin{cases} E_1 = \epsilon - \Delta \\ E_2 = \epsilon + \Delta \end{cases}$$ **Eigen value**

Put $E_1 = E - \Delta$ **back:** $\begin{pmatrix} \Delta & \Delta \\ \Delta & \Delta \end{pmatrix}$ **and the algebraic cofactor is** $\begin{pmatrix} \Delta & -\Delta \\ -\Delta & \Delta \end{pmatrix}$

The adjoint matrix is the transpose of the cofactor: $\begin{pmatrix} \Delta & -\Delta \\ -\Delta & \Delta \end{pmatrix}$

III. What is the eigenstate and eigenvalue?

According to linear algebra, any non-zero column vector of the adjugate matrix $adj(\hat{O})$ is directly an eigenvector corresponding to that eigenvalue

$$\begin{pmatrix} \Delta & -\Delta \\ -\Delta & \Delta \end{pmatrix}$$

Eigen state $\boldsymbol{\phi_1} = \begin{pmatrix} \Delta \\ -\Delta \end{pmatrix}$

Normalisation $\sqrt{(\Delta)^2 + (-\Delta)^2} = \sqrt{2}\Delta = 1$

$$\boldsymbol{\phi_1} = \frac{1}{\sqrt{2}\Delta}\begin{pmatrix} \Delta \\ -\Delta \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\boldsymbol{\phi_2} = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Eigen value $\begin{cases} E_1 = \epsilon - \Delta \\ E_2 = \epsilon + \Delta \end{cases}$ **Eigen state** $\boldsymbol{\phi_i} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$

IV. What basis is used for the wavefunction expansion?

Localized Basis — Linear Combination of Atomic Orbital (LCAO) basis

The wavefunction $|\Psi\rangle$ is expanded using a set of atom-centered, localized orbitals

$$\Psi(r) = \sum_i \sum_\mu c_{i\mu} \phi_\mu (r - R_i)$$

atomic orbital

position of *i*-th nucleus

Form of ϕ_μ

- **Slater-Type Orbitals (STOs)** $\phi_{STO}(r) = N \cdot r^{n-1} e^{-\zeta r} Y_{lm}(\theta, \phi)$
- **Gaussian-Type Orbitals (GTOs)** $\phi_{GTO}(r) = N \cdot x^a y^b z^c e^{-\alpha r^2}$

where, N is the normalization constant, $Y_{lm}(\theta, \phi)$ represents the spherical harmonics, and ζ, α are orbital exponents governing the spatial decay of the electron density

IV. What basis is used for the wavefunction expansion?

Plane Wave Basis Sets

For periodic systems (like crystals), Bloch's theorem states that the electronic wavefunction can be expanded using a discrete set of plane waves that match the periodicity of the reciprocal lattice

$$\Psi_k(r) = \sum_G c_{k,G}\phi_G(r) \quad \text{and} \quad \phi_G(r) = \frac{1}{\sqrt{\Omega}} e^{i(k+G)\cdot r}$$

***k* is the crystal momentum vector within the first Brillouin zone**

***G* is the reciprocal lattice vector satisfying $e^{iG\cdot R} = 1$ (where *R* is any real-space lattice vector)**

***Ω* is the volume of the unit cell acting as a normalization factor**

Kinetic Energy Truncation (E_{cut}): To make the matrix finite, the summation is restricted to a sphere in reciprocal space defined by the kinetic energy cutoff

$$\frac{\hbar^2}{2m}|k+G|^2 \leq E_{cut}$$

IV. What basis is used for the wavefunction expansion?

Real-Space Grid Basis Sets (Finite Difference)

In a real-space grid approach, we bypass analytical functions entirely. The basis functions are mathematical delta functions located at discrete mesh coordinates

The Coordinate Definition

In a 1D domain discrete grid with uniform spacing Δx: $x_j = x_0 + j \cdot \Delta x \quad (j = 1, 2, 3, \ldots, N)$

The basis functions are effectively a set of orthogonal Kronecker delta functions centered at each grid point x_j

$$\phi_j(x) = \delta(x - x_j)$$

IV. What basis is used for the wavefunction expansion?

The continuous wavefunction $\Psi(x)$ is represented as a linear combination of these localized grid spikes

$$\Psi(x) \approx \sum_{j=1}^{N} \Psi_j \cdot \boldsymbol{\phi_j}(x)$$

When written in the discrete Dirac representation, the expansion coefficients are simply the values of the wavefunction at those exact grid coordinates, forming a direct column vector for computer memory allocation

$$|\Psi\rangle = \begin{pmatrix} \Psi(x_1) \\ \vdots \\ \Psi(x_n) \end{pmatrix}$$



That's all for this session

Thank you for your attention and great participation